

# KalingApp Backend — Automated Test Suite

47 tests, all passing · 100% coverage on the booking state machine and Smart Allocation ranking logic · 82% combined coverage across the accounts and milkbank apps · generated 2026-09-05, backend @ commit 87603ce

## accounts/tests.py

### RegistrationAndVerificationTests — *The full register -> email-verify -> login lifecycle.*

|                                                                         |                                                                                          |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <code>test_register_creates_inactive_account_and_emails_a_code</code>   | Registering creates the account as inactive and emails a real 6-digit verification code. |
| <code>test_register_defaults_blank_baby_name_to_james</code>            | Leaving the baby's name blank at registration defaults it to "James."                    |
| <code>test_register_rejects_duplicate_email</code>                      | Registering with an email that's already in use is rejected.                             |
| <code>test_cannot_log_in_before_verifying</code>                        | Logging in before verifying the email fails, the same way a wrong password would.        |
| <code>test_verify_with_correct_code_activates_and_returns_tokens</code> | The correct verification code activates the account and returns real login tokens.       |
| <code>test_verify_with_wrong_code_increments_attempts_and_fails</code>  | An incorrect code fails and counts against the attempt limit.                            |
| <code>test_verify_locks_out_after_max_attempts</code>                   | Too many wrong codes blocks further attempts, even once the real code is tried.          |
| <code>test_verify_rejects_expired_code</code>                           | A code older than its 15-minute window is rejected.                                      |
| <code>test_resend_respects_cooldown</code>                              | Resending a code too soon is a no-op until the cooldown period passes.                   |
| <code>test_resend_after_cooldown_issues_a_new_code</code>               | After the cooldown, resending issues a genuinely different code.                         |
| <code>test_resend_does_not_revealWhether_an_email_exists</code>         | Resending for an unregistered email returns the same generic response either way.        |

### ProfileAndConsentTests — *Profile access, editing, and GPS consent.*

|                                              |                                                             |
|----------------------------------------------|-------------------------------------------------------------|
| <code>test_me_requires_authentication</code> | /auth/me/ refuses to respond without a logged-in session.   |
| <code>test_me_returns_own_profile</code>     | A logged-in user receives back their own real profile data. |

|                                                                  |                                                                                                            |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <code>test_patch_me_updates_allowed_fields_only</code>           | Editing a profile only changes fields it's allowed to — <code>is_staff</code> /email can't be smuggled in. |
| <code>test_location_consent_requires_consent_true</code>         | Submitting a location without explicit consent is rejected.                                                |
| <code>test_location_consent_stores_coordinates_when_given</code> | Location is saved once consent is explicitly given.                                                        |

## CheckInStreakTests — *The three-way branch behind the breastfeeding streak counter.*

|                                                                    |                                                                |
|--------------------------------------------------------------------|----------------------------------------------------------------|
| <code>test_first_ever_checkin_sets_streak_to_one</code>            | A first-ever check-in sets the streak to 1.                    |
| <code>test_checking_in_twice_same_day_does_not_double_count</code> | Checking in twice on the same day does not inflate the streak. |
| <code>test_checking_in_next_day_increments_streak</code>           | Checking in exactly one day later increases the streak by 1.   |
| <code>test_checking_in_after_a_gap_resets_to_one</code>            | Checking in after skipping a day resets the streak back to 1.  |

## FacilityStaffPermissionTests — *Making sure `is_staff` and `facility_staff` never get confused.*

|                                                                              |                                                                                                                 |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>test_mother_cannot_list_staff_user_management_endpoint</code>          | A mother account is blocked from the facility staff's user-management list.                                     |
| <code>test_facility_staff_can_list_users</code>                              | A real <code>facility_staff</code> account can access the user-management list.                                 |
| <code>test_django_is_staff_alone_does_not_grant_facility_staff_access</code> | A Django admin ( <code>is_staff</code> ) account without the <code>facility_staff</code> role is still blocked. |

## milkbank/tests.py

### TransitionsTests — *`apply_transition()` — the one function allowed to change a booking's status.*

|                                                                     |                                                                                           |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <code>test_every_status_has_an_entry_in_the_transition_table</code> | Every booking status has a defined set of allowed next moves — none are silently missing. |
| <code>test_valid_transition_succeeds_and_is_audited</code>          | A legal status change goes through and updates the booking.                               |

|                                                                              |                                                                                             |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <code>test_valid_transition_notifies_the_owner</code>                        | A status change sends the mother the correct notification message.                          |
| <code>test_invalid_transition_is_rejected_and_leaves_status_unchanged</code> | An illegal status jump (e.g. declined → completed) is blocked; the booking stays as it was. |
| <code>test_pending_cannot_jump_straight_to_scheduled</code>                  | A pending booking can't skip straight to scheduled without going through acceptance first.  |
| <code>test_terminal_status_decrements_facility_booked_count</code>           | Declining, expiring, or completing a booking frees up the facility's slot count.            |
| <code>test_non_terminal_status_does_not_touch_booked_count</code>            | A non-final status change leaves the facility's slot count untouched.                       |
| <code>test_completing_a_donor_request_creates_a_donation_record</code>       | Completing a donor's booking creates a “Donation” transaction record.                       |
| <code>test_completing_a_recipient_request_creates_a_received_record</code>   | Completing a recipient's booking creates a “Received” transaction record.                   |
| <code>test_counter_offer_can_return_to_pending_or_go_to_scheduled</code>     | A countered booking can legally return to pending after the mother rejects it.              |

## SmartAllocationRankingTests — *The facility-matching tie-break chain, tested as pure functions.*

|                                                                           |                                                                                            |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <code>test_lower_booked_ratio_wins_regardless_of_raw_count</code>         | A facility with a lower booked-to-capacity ratio ranks first, even with more raw bookings. |
| <code>test_donor_prefers_lower_stock_facility_on_ratio_tie</code>         | On a ratio tie, a donor is matched to whichever facility needs milk the most.              |
| <code>test_recipient_prefers_higher_stock_facility_on_ratio_tie</code>    | On a ratio tie, a recipient is matched to whichever facility has the most milk to give.    |
| <code>test_distance_breaks_ties_when_ratio_and_stock_are_equal</code>     | Distance only decides the ranking after ratio and stock are already tied.                  |
| <code>test_ranked_facilities_carry_the_computed_ratio_and_distance</code> | Each ranked facility carries its calculated ratio and distance for the client to display.  |

## GetRankedFacilitiesTests — *The eligibility gates that run before ranking even starts.*

|                                                                 |                                                                  |
|-----------------------------------------------------------------|------------------------------------------------------------------|
| <code>test_raises_location_required_with_out_coordinates</code> | Ranking fails clearly if the mother has no GPS location on file. |
|-----------------------------------------------------------------|------------------------------------------------------------------|

|                                                                     |                                                                                           |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <code>test_raises_no_operational_facility_when_none_match</code>    | Ranking fails clearly if no facility is operational or has open capacity.                 |
| <code>test_recipient_excludes_facilities_below_minimum_stock</code> | A recipient is never matched to a facility below the minimum stock cutoff.                |
| <code>test_donor_is_never_excluded_by_low_stock</code>              | A donor can still be matched to a low-stock facility — exactly who needs a donation most. |

## BookingEndpointPermissionTests — *Who's allowed to call which booking action, over real HTTP requests.*

|                                                                              |                                                                          |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| <code>test_mother_cannot_call_staff_accept</code>                            | A mother account is blocked from the staff-only “accept booking” action. |
| <code>test_staff_can_call_staff_accept</code>                                | A real facility_staff account can accept a booking.                      |
| <code>test_a_different_mother_cannot_confirm_someone_elses_attendance</code> | One mother can't confirm attendance on a different mother's booking.     |
| <code>test_owner_can_confirm_their_own_attendance</code>                     | A mother can confirm attendance on her own booking.                      |
| <code>test_cannot_have_two_open_requests_at_once</code>                      | A mother can't submit a second booking while one is already open.        |

Total: 47 tests. Run with `python manage.py test` from the backend project root. Every other app (articles, notifications, chat, directory, core, dashboard) and all three frontends currently have zero automated tests.